Digital crew intercom for an armoured vehicle: engineering dossier

Rev A, 9 Nov 2026. Portfolio and thesis-preparation work; not a product, not certified.

This dossier follows the systems-engineering spine of Wasson, *System Engineering Analysis, Design, and Development*, 2nd ed. (Wiley, 2016) [W]. Each section names the chapter and pages it applies. Every engineering number carries a reference key [Rn] (list in §16) and a status: - **V**: read in the source. - **G**: read off a published graph, ±2 dB. - **D**: derived from V/G values by a stated method. - **A**: an assumption of this study.

The models, firmware and tests that produce the results are in the same repository. §14 lists what is assumed and what is sourced.

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0. The question and the short answer

Open engineering question. Can crew speech stay intelligible in vehicle noise of 110 dB(A) or more, within a fixed latency and sidetone budget and within the hearing-exposure limit? And how does the intercom stay fail-operational when a crew box, the bus or the central unit fails?

Short answer, from the models in this repository:

1. **Intelligibility is lost on the transmit side, not at the ear.** - In the 111 dB(A) tracked-vehicle spectrum [R2], a pressure-gradient boom microphone at 1.5 cm with a "raised" voice (the MIL-STD-1474E prediction assumption [R4]) gives a transmit SNR of –4 to –8 dB in the 2–8 kHz bands. - That caps the STI at the listener at **0.44**, whatever the earphone level, the ANR or the codec. - The threshold STI 0.41 (MRT 85 %, MIL-STD-1474E 4.3.1 and Table I [R4]) is reachable. - The objective 0.48 (MRT 90 %) needs **at least 3 dB more transmit SNR**. That comes from the microphone and its noise shield, or the talker's effort, not from DSP (S1, §9.2).
2. **The hearing dose is the binding constraint at 110 dB(A) and above.** - Under the crew helmet alone [R17] the ear sees 94.8 dB(A). - Broadband ANR [R3] brings this to 85.9 dB(A), and the speed-referenced tonal ANR of this design takes a further 0.6 dB off. - Even then, the 8-h TWA at the operating point is 89.9 dB(A), against the 85 dB(A) limit of MIL-STD-1474E 4.2.3.1 c [R4]. That allows 2.6 h at the ear. - The configurations studied stay within dose and intelligibility up to **108–109 dB(A)** (tracked) and **105 dB(A)** (wheeled). - Above that, about **10 dB more passive attenuation** is needed (a communication earplug under the helmet, A). The digital design cannot supply it. - With the earplug and a 3 dB better microphone (C6), the limit moves to 114 dB(A) tracked and 113 dB(A) wheeled, at STI 0.55.
3. **Latency decides where each kind of ANR may run** (S2, §9.3). - Broadband feedback ANR is bounded by how predictable the noise is over the loop delay:
    - 34 dB at 50 µs;
    - 12 dB at 0.5 ms;
    - about 1 dB at 2 ms.
    - So it must stay analog or sample-by-sample on a low-latency converter path.
    - Tonal ANR referenced to the track-laying frequency tolerates the whole 1 ms frame pipeline. On speed-jittered tones it keeps 57 dB at 0.2 ms, 46 dB at 2 ms and 35 dB at 8 ms of loop delay.
    - It is only as good as its reference, though:
    - a **sprocket-phase pick-up** gives about 35 dB on jittering track tones;
    - the bus speed value at 50 Hz gives 29 dB;
    - a 1 % speed-scale error drops it to 18 dB at best;
    - 0.5 Hz of tone wander that no sensor sees drops it to 10 dB at best.
4. **Fail-operational is achievable and cheap with an analog emergency path**, but only as good as fault detection (S6, §10). - A digital star without a bypass loses commander–driver voice in 693 of 10^6 12-h missions. - With a fail-safe relay to an analog party line, that falls to 37 per 10^6. - At 50 % BIT coverage of internal crew-box faults it rises to 183 per 10^6. - The crew-box firmware drops to the bypass 20 ms after the last bus frame, or within about 2 ms (1 tick + the relay hold time) if the processor stalls. Both are measured in Renode (§13.3).
5. **Architecture.** In the weighted analysis of alternatives, the digital star with CCU mixing and an analog bypass (A2) scores 7.47 and TSN with distributed mixing (A3) scores 7.38. A2 wins 68 % of 20 000 random weightings, and no ±10 % single-weight change flips the result (S5, §8).

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1. Problem space (W ch. 4, pp. 76–98)

Problem statement (one sentence, no solution words; W p. 92). Crew members cannot reliably understand each other or

the radio nets while the vehicle moves at high noise levels, or after a single equipment failure.

Discussion. - Interior noise of tracked and wheeled armoured vehicles at speed is 110–125 dB(A). - A tracked APC measured 113–117 dB(A) at 40 km/h [R1] (V). - A tracked and a wheeled armoured vehicle measured 111 and 110 dB(A) at 60 km/h [R2] (V). - A NATO survey of 26 land vehicles gives 80 dB(A) to above 110 dB(A) [R3] (V). - MIL-STD-1474E puts any area at 100 dB(A) or more into category A: no direct person-to-person communication is expected. Below that, category B needs "electrically aided communication via an attenuating helmet or headset" [R4 Table A-I] (V). - The intercom is therefore the only voice path between crew members. Its loss in a tactical phase is a mission and safety hazard.

Need typing (W Principle 4.12, p. 89). - **Real need:** intelligible crew coordination at full noise. - **Projected need:** more radio nets and data on the same harness. - **Perceived need:** "louder headsets". Loudness is not the need (§9.2), and it costs noise dose.

Stakeholders (after W Table 4.2, pp. 84–85).

Role	Description	Intercom concern
Commander (user)	directs crew, works 1–3 radio nets	override priority, monitoring several nets
Driver, gunner, loader (users)	execute orders	intelligibility at full noise, hands-free
Higher command (end user)	on the radio net	radio path latency, rebroadcast
Maintainer	field LRU swap	BIT localises the fault to a crew box, cable or headset
Vehicle integrator	power, harness, EMC	28 V transients, harness mass, emissions
Acquirer / test authority	buys, qualifies	MIL-STD-461/810/1275 evidence, MRT/STI test
Threat	EW, acoustic signature	emission control, no radiated audio

Problem and solution space (W Principles 4.17–4.18, Fig. 4.7, pp. 93–95). - Problem space 1.0 "crew speech communication" splits into these solution spaces, each then a problem space of its own: - 1.1 headset (mic, earphone, passive and ANR); - 1.2 crew box; - 1.3 central communication unit (CCU); - 1.4 intercom bus and harness; - 1.5 radio interfaces; - 1.6 power. - The question of §0 is a **critical technical issue** (CTI, W p. 100): STI, latency and noise dose must be met by the same design.

2. Mission analysis, use cases and scenarios (W ch. 5, pp. 99–128)

Mission statement (one outcome, W Principle 5.2): the crew keeps intelligible voice coordination, internally and with external nets, for the whole vehicle mission.

Mission objectives. - **Performance:** STI at the listener of at least 0.41 (threshold) and 0.48 (objective) in the operating noise. - **Mission reliability:** commander–driver voice for a 12-h operating period (A) with a probability of at least 0.9999. - **Constraint:** noise exposure at the ear within MIL-STD-1474E.

Phases (W Principle 5.7, p. 110): - **Pre-mission:** power-up, BIT, load the net plan, fit the headset. - **Mission:** road march, tactical move, engagement, recovery. - **Post-mission:** log download, headset care.

Mission event timeline (W Fig. 5.5, p. 112):

```
t0 vehicle power on -> t1 intercom ready (BIT pass, all boxes NORMAL) -> t2 nets keyed
-> t3 movement (noise 100-115 dB(A)) -> t4 contact / engagement (noise peaks, gun) -> t5 damage event
   (box, cable or CCU lost: emergency path within 20 ms + relay time) -> t6 recovery -> t7 power off
```

MOE/MOP tree (W Fig. 5.8, p. 116). - **MOE:** intelligible crew coordination. - **MOPs:** - STI at the listener; - mouth-to-ear latency; - sidetone delay; - ANR attenuation; - failover time to the emergency path; - noise dose at the ear. - **MOSs:** - crew-box MTBF; - LRU swap time; - BIT coverage; - operation with gloves.

Use cases (verb–outcome titles; priority = frequency × criticality, W Principle 5.18, p. 121).

ID	Use case	Primary actor	Frequency	Criticality
UC1	Talk to the crew	any crew member	continuous	high
UC2	Talk on a radio net	commander	frequent	high
UC3	Monitor several nets with priority	commander	frequent	medium
UC4	Hear a vehicle warning over all audio	vehicle	rare	high
UC5	Keep talking after a crew-box, bus or CCU failure	any crew member	rare	**high**
UC6	Rebroadcast between two radio nets	commander	occasional	medium
UC7	Maintain and calibrate the ANR at standstill	maintainer	rare	low

UC5 actor–system exchange (after W Table 5.1, p. 123):

Step	User (actor)	System (actor)
1	talks normally	carries speech through the crew box, bus, CCU, bus and crew box (NORMAL)
2	–	the CCU dies, so downlink frames stop
3	–	each crew box conceals 3 frames, then mutes; after 20 ms without a frame it de-energises its relay (BYPASS)
4	keeps talking	the headset is now on the analog emergency party line; every box in BYPASS hears every other
5	–	the box keeps sending telemetry-only frames; when the link is good for 2 s, it returns to NORMAL

Scenarios and compensating provisions (W Principle 5.23, pp. 122–125):

Scenario	Compensating provision
CCU dead	analog bypass in every box
bus cut at one box	that box goes to BYPASS; procedure: the commander orders "all on emergency"
processor hang	dynamic relay drive drops the relay
wrong ANR calibration	divergence guard switches the ANR off
28 V surge or crank	input stage design (§11.3)
headset fault	BIT, plus a spare headset

3. Concept of operations (W ch. 6, pp. 129–146)

Operations (W Table 6.4, p. 143). Normal, degraded and emergency operations are specified separately, not as footnotes: - **Normal ops:** digital path. Each crew box has local sidetone and ANR. The CCU mixes intercom and radio channels and sends four channels to each box; each box applies priority and volume locally. - **Degraded ops:** one function lost (the ANR off after its guard trips). The digital path continues. - **Emergency ops:** the CCU or a link is lost, or a crew box has a critical fault. That box's headset is switched by a fail-safe relay onto an independent analog emergency line.

Maintenance concept. - The crew box, CCU and headset are LRUs, swapped at field level (MTTR 0.5 h, A). - BIT localises a fault to the LRU and reports through the uplink telemetry.

Context diagram (Level 0):

```
            28 V vehicle power (MIL-STD-1275E)      vehicle data bus (speed, sprocket phase)
                          |                                    |
radios 1..n <--audio/PTT--> [ CCU ] <==bus==> [ crew box 1 ] <--> headset 1 (mic, earphones, ANR mic)
                            |     <==bus==> [ crew box 2 ] <--> headset 2
                            |     <==bus==> [ crew box n ] <--> headset n
                            +----- analog emergency party line (all crew boxes, independent of CCU) ----
environment: 110–125 dB(A) noise, shock/vibration, EMI from own transmitters, −46..+71 C
```

4. Phases, modes and states (W ch. 7, pp. 147–173)

Vocabulary (W p. 148; Principles 7.1–7.7): - **Phases** are the mission phases of §2. - **Modes** are selected by the user or the system: - CCU system modes: NORMAL, REBROADCAST, MAINTENANCE. - Crew-box operating states: INIT, NORMAL, DEGRADED, BYPASS, MAINT. - **Configuration states:** relay energised or de-energised, ANR on or off, net routing. - **Environmental states:** noise band (<85, 85–100, 100–110, ≥110 dB(A)). - **Entry into DEGRADED and BYPASS is automatic**, triggered by a health change. Wasson lists "Degraded Operations" both as a mode (Table 7.3, p. 162) and as an operational state (p. 156). This dossier treats it as an automatic transition with its own allowable actions.

Crew-box transition table (W Table 7.1 format, p. 158, with the time constraint of Fig. 7.6; exit criteria of the current state, W Author's Note 7.4, p. 159). It is implemented in `fw/lib/icom/src/modes.c` and tested in `fw/tests/unit/test_modes.c` and in Renode IT01–IT08.

Current	Event	Trigger (criteria)	Next	Time limit	Req.
INIT	E1	first valid downlink frame, no critical BIT fault	NORMAL (DEGRADED if ANR fault)	next 1 ms tick	CB-20
INIT	E2	no link within 2 s of POST	BYPASS (not latched)	2 s	CB-21
any	E3	critical BIT fault (codec, supply)	BYPASS, **latched**	next tick	CB-24
NORMAL, DEGRADED, MAINT	E4	no valid frame for 20 consecutive ticks	BYPASS	20 ms + 1 tick	CB-22
BYPASS (not latched)	E5	2000 consecutive good frames	NORMAL / DEGRADED	2 s	CB-23
NORMAL	E6	ANR guard trip	DEGRADED	0.2 s after the trip criterion	CB-25
DEGRADED	E7	ANR fault cleared (reset)	NORMAL	next tick	CB-25

NORMAL, DEGRADED	E8	MAINT_ENTER command **and speed = 0**	MAINT	next tick	CB-28
MAINT	E9	MAINT_EXIT, speed > 0, or 10 min	NORMAL / DEGRADED	next tick	CB-28
any digital	E10	processor stall (no relay-drive toggle)	relay drops (hardware)	1 tick + relay hold	CB-27

The 2 s recovery hysteresis (E5) is deliberate. A link that keeps dropping must not toggle the crew between the digital path and the emergency line. Renode IT03 holds the link at 50 % frame loss for 3 s and sees zero transitions.

Allowable/prohibited actions (W Fig. 7.14, p. 171). P cells flow down as design constraints (W p. 172).

Mode \ action	talk crew	talk radio	change routing	rotate S_hat / calibrate ANR	firmware update
NORMAL	A	A	A	**P** (refused, `box.c command()`)	P
DEGRADED	A	A	A	P	P
BYPASS	A (analog line)	P (no radio on the emergency line, A)	P	P	P
MAINT (standstill only)	A	A	A	A	A

CCU system modes. - REBROADCAST connects radio 1 receive to radio 2 transmit and the reverse, with the commander monitoring. It is a CCU routing state; the crew box needs no change because it receives channels with activity flags. - The CCU itself is outside the firmware built here (§14).

5. Requirements (W ch. 19–22, pp. 397–464)

Approach. - Performance-based at system level; the block diagram stays an internal model, not a mandated design (W Principle 20.5, p. 424). - One measure of performance per requirement (W Principle 20.1, p. 418). - Each statement has subject, "shall", verb, outcome, level and condition (W Principle 22.5, p. 449). - Threshold and objective are tagged inline (W p. 410). - The wording rules ban "all", "etc." and "and/or" (W Principle 22.21, p. 461).

System requirements derived from standards (outline after W Table 20.1, pp. 417–418; operations categories after W Fig. 19.5 and Principle 19.22, pp. 408–411):

ID	Requirement (abridged)	Source (trace)	Verif.
SYS-3.2.1	In the noise of §3.5 at 110 dB(A), the intercom shall deliver crew speech at an STI ≥ 0.41 (T) / ≥ 0.48 (O), measured per IEC 60268-16 ed. 5.	MIL-STD-1474E 4.3, 4.3.1 and Table I (MRT 85 % → STI 0.41) [R4]; MIL-STD-1472G Table XX (91 % MRT normal acceptable) [R5]	A (S1), T
SYS-3.2.2	The intercom shall keep the 8-h TWA at the ear below 85 dB(A), including the intercom output.	MIL-STD-1474E 4.2.3.1 c [R4]	A (S1), T
SYS-3.2.3	The earphone output shall not exceed 115 dB peak at the ear.	MIL-STD-1472G 5.3.1.8.2 a(3) [R5]	T
SYS-3.2.4	Sidetone delay shall be ≤ 10 ms (design-to; the human limit is 20 ms).	Stone & Moore 1999/2002, "disturbing" at 20–30 ms [R20][R21]	A (S4), T
SYS-3.2.5	Crew-to-crew mouth-to-ear delay shall be ≤ 20 ms.	proxy: own-voice limit [R20][R21] (A); G.114 150 ms is the end-to-end bound [R24]	A, T
SYS-3.2.6	The binaural delay between the two earphones shall be ≤ 1 ms.	MIL-STD-1472G 5.3.1.6 i [R5]	I
SYS-3.2.7	After any single failure of a crew box, a bus link or the CCU, commander–driver speech shall resume within 50 ms.	W Fig. 19.5 abnormal and emergency ops; UC5	T (Renode IT02/IT08), A
SYS-3.2.8	Mission reliability of commander–driver speech over 12 h shall be ≥ 0.9999.	mission objective §2 (A)	A (S6)
SYS-3.3.1	Vehicle warnings shall be heard over every other channel.	UC4	T
SYS-3.3.2	Sidetone shall be provided in phase, unfiltered.	MIL-STD-1472G 5.3.1.6 j [R5]	I, T
SYS-3.3.3	The microphone path shall improve peak-speech-to-noise by ≥ 10 dB over a non-noise-cancelling microphone in ≥ 100 dB low-frequency noise.	MIL-STD-1472G 5.3.1.6 d(2) [R5]	T
SYS-3.4.1	The intercom shall operate from 20 to 33 V and ride through 1275E surges, spikes, IES and cranking without reset.	MIL-STD-1275E 5.1.1–5.1.3, Table I [R27]	A (S7), T
SYS-3.5.1	The equipment shall meet MIL-STD-461G CE102, CS101, CS114, CS115, CS116, RE102 and RS103 (Army ground).	MIL-STD-461G Table V [R29]	T
SYS-3.5.2	The equipment shall meet the MIL-STD-810H methods of §11.2.	MIL-STD-810H [R30]–[R32]	T

Flow-down (W ch. 21, pp. 429–445). - The UC1 thread (capture → crew box → bus → CCU → bus → crew box →

earphone) apportions the latency budget (§7.2). STI is a synthesis budget, flowed down through the validated model (W Principle 31.5). - The crew-box software requirements are in [fw/SRS.md](): CB-01 … CB-47 with lineage to the SYS IDs. `fw/tools/trace.py` checks every CB requirement against a unit test or a Renode test, and fails CI otherwise.

Allocation matrix (W Fig. 21.9, p. 440). X marks where a requirement is allocated.

Capability	Headset	Crew box HW	Crew-box SW	Bus	CCU	Procedures
STI (SYS-3.2.1)	X (mic, passive)	X (codec)	X (gain, limiter)		X (mix)	X (voice effort)
Noise dose (SYS-3.2.2)	X (passive, ANR)		X (limiter, tonal ANR)			X (mission time)
Latency (SYS-3.2.4/5)		X (converters)	X (frame)	X	X	
Fail-operational (SYS-3.2.7)		X (relay, emergency amp)	X (supervision, relay drive)	X	X	X (drill)

6. Architecture and interfaces (W ch. 26–27, pp. 542–598)

Level 1 architecture (selected A2, §8). - **Crew box** (one per station; RT1064-class MCU, audio codec, RS-485 transceiver, relay, emergency amplifier, controls). - **CCU** (radio interfaces, mixing, routing, system modes). - **Star bus:** point-to-point RS-485, 1 ms TDM frames. - **Analog emergency party line,** independent of the CCU. - **Headsets** with a passive helmet, broadband ANR, an error microphone and a pressure-gradient boom microphone.

Redundancy types (W pp. 565–568): - The emergency path is **cold-standby, unlike redundancy**: analog against digital, so the two do not share common-mode software or clock faults. - It is engaged by a fail-safe relay: de-energised means bypass. - The relay coil is powered through a charge pump driven by a toggling GPIO, so a stalled processor drops the relay without software help. A static GPIO level would not. - A single connector into both paths would still be a single failure point (W Fig. 26.11, p. 567), so the emergency line uses its own connector pins (A).

Control (W Fig. 26.9, pp. 562–564). - Mixing is centralised in the CCU. Priority, volume, sidetone, ANR and failure detection are distributed to each crew box, so a CCU loss only removes mixing and radios. - The TSN alternative (A3) would also distribute mixing (§8).

Interface register (W Principle 27.6, keyed source–destination):

ID	From → to	Type	Content
HS-CB	headset → crew box	analog	boom mic (balanced), error mic, PTT
CB-HS	crew box → headset	analog	earphone drive (speech + sidetone + anti-noise)
CB-BUS / BUS-CB	crew box ↔ CCU	RS-485, 1 ms TDM frames	uplink: state, BIT, telemetry, mic audio; downlink: 4 audio channels, flags, speed, command
CB-EL	crew box ↔ emergency line	analog, transformer-isolated	party line
VPWR	vehicle → crew box, CCU	28 V DC	MIL-STD-1275E envelope
VDB	vehicle → CCU	data	speed and sprocket phase (for the tonal ANR)

Bus frame ICD (after the packet template of W Figs. 27.4/27.5, p. 590; implemented in `fw/lib/icom/src/frame.c`).

```
0xA5 0x5A | len | type | payload[len] | CRC-16/CCITT-FALSE (over len, type, payload)
```

- **Downlink** (type 0x01):
- seq, CCU mode, channel-active flags;
- speed (0.01 km/h);
- cmd, arg;
- nch;
- nch × 16 samples × int16 at 16 kHz.
- **Uplink** (type 0x81):
- seq, box ID, state, BIT bits, PTT;
- telemetry ID, telemetry value (rotating: state/cause, ANR attenuation, frames lost, CRC errors);
- nch;
- 0 or 16 samples.
- **Size and load:**
- a full downlink is 143 bytes per ms, which is 1.43 Mbit/s with 8N1;
- at 10 Mbit/s RS-485 that is **14 % of the link**, inside the 50 % utilisation goal (W p. 590);
- the uplink is 47 bytes per ms (4.7 %).

7. Performance budgets and margins (W ch. 31, pp. 666–681)

Method: W Fig. 31.1 and Example 31.3 (pp. 669–671). Each budget has an allocated value, a design-to value and a margin. The reserve is held at the top level, with no stacked 10 % per leaf (p. 671). Worst-case (late-finish) values are used (p. 674).

7.1 STI and noise-dose budget (synthesis, through the S1 model)

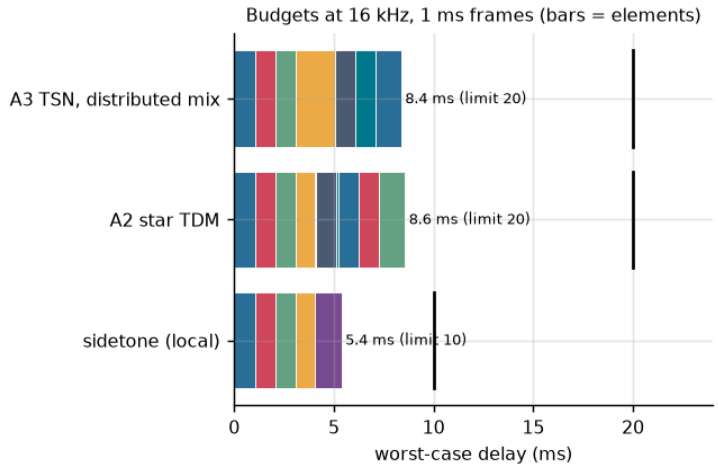
Tracked vehicle, 111.3 dB(A) and 124.9 dB lin at the crew position [R2]; raised voice [R4]; mic at 1.5 cm (A).

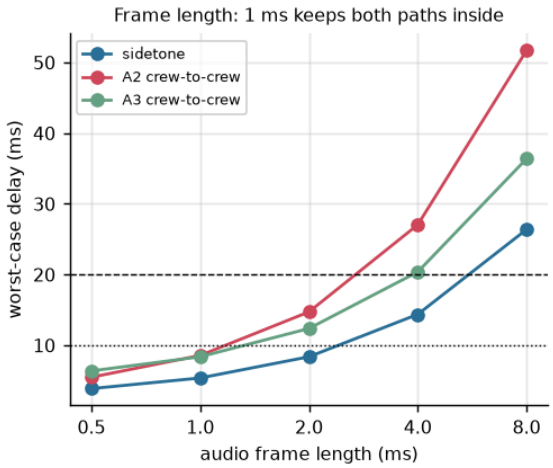
Item	Value	Basis
Ear noise, helmet only	94.8 dB(A)	[R2] – helmet attenuation [R17]
Ear noise, + broadband ANR	85.9 dB(A)	– ANR insertion gain [R3] (G)
Ear noise, + tonal ANR 20 dB on tonal share	85.3 dB(A)	tonal share from the 1/3-octave spectrum (D): 94 % / 58 % / 50 % in 63/125/250 Hz
Transmit SNR at the mic, 125 Hz … 8 kHz	–0.4, 2.3, 9.8, 5.3, –3.9, –5.9, –7.6 dB	ANSI S3.5 raised speech [R14] + 36.5 dB near field (A) vs cab noise – mic rejection [R18]
Best STI, C3 (wideband, both ANRs)	**0.44**	IEC 60268-16 indirect [R11][R13], validated by STIPA (§9.4)
Speech level at the ear for STI 0.41	91 dB(A)	S1
8-h TWA at that point (speech duty 30 %, A)	89.9 dB(A)	limit 85: **–4.9 dB margin (fails)**; allowed 2.6 h
Transmit SNR needed for the objective 0.48	+3 dB	C5: STI 0.52, TWA 87.0
Passive attenuation needed for 8 h at 111 dB(A)	+10 dB (earplug, A)	C6: ear noise 75.3 dB(A), STI 0.55, TWA 77.0 dB(A); limit of use 114.3 dB(A)

7.2 Latency budgets (apportionment along the UC1 thread)

16 kHz sampling, 1 ms frames, converter filter A: ADC 17/fs, DAC 21/fs [R37] (V).

Path	Elements (ms)	Total	Limit (design-to)	Margin
Sidetone, local in the crew box	ADC 1.06, input frame 1, processing 1, output frame 1, DAC 1.31	**5.4 ms**	10 ms	4.6 ms (46 %)
Crew-to-crew, A2 star TDM	ADC 1.06, frame 1, box 1, slot wait 1, uplink 0.05, CCU mix 1, downlink 0.14, concealment buffer 1, output 1, DAC 1.31	**8.6 ms**	20 ms	11.4 ms (57 %)
Crew-to-crew, A3 TSN	ADC 1.06, frame 1, box + packet 1, network 2 (class A bound [R36]), mix 1, output 1, DAC 1.31	**8.4 ms**	20 ms	11.6 ms (58 %)
Radio path through the box	as A2	8.6 ms	10 ms of the 150 ms G.114 budget [R24] (A)	1.4 ms





Frame-length sensitivity (S4). Sidetone takes 3.9 / 5.4 / 8.4 / 14.4 ms at 0.5 / 1 / 2 / 4 ms frames. A 2 ms frame would leave only 1.6 ms of sidetone margin and 4 ms breaks it. **The 1 ms frame is a derived requirement (CB-02).**

MIL-STD-1472G 5.3.1.6 j asks for sidetone "in phase". A digital path cannot be truly in phase. The 5.4 ms digital sidetone meets the human-factors limit, but the clean solution is the codec's analog microphone-to-headphone bypass. This is recorded as an open design decision (§15).

7.3 Processing budget

One `icom_box_tick` per 1 ms frame: 16 samples × 3 harmonics of FxLMS, mixer, link and modes. The worst case measured in Renode is in §13.3, against the 1 ms frame (a design-to limit of 0.5 ms keeps 50 % for the codec DMA, the bus and later functions; A). Renode executes instructions at a configured rate, not real Cortex-M7 timing, so this is a sanity check. A WCET needs the real board.

8. Analysis of alternatives (W ch. 32, pp. 682–702)

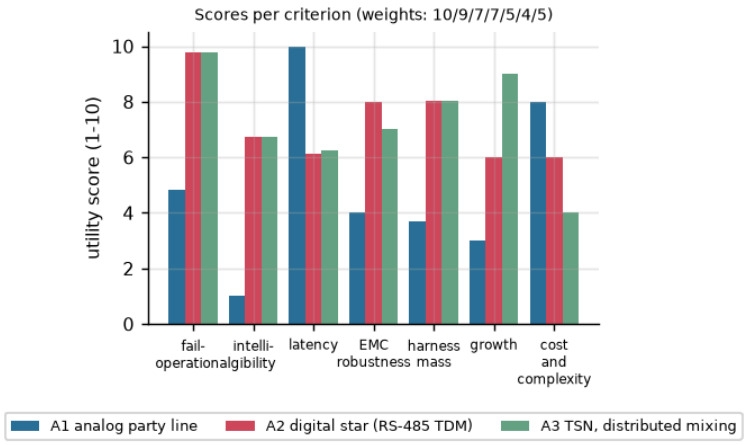
Trade-study report in the outline of W Table 32.1 (p. 691), condensed.

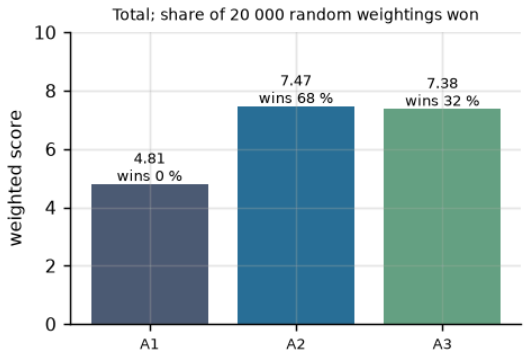
Scope. Choose the intercom transport and mixing architecture.

Trade space (W Fig. 32.4). - A1 analog party line. - A2 digital star: RS-485 point-to-point TDM, CCU mixing, local priority, sidetone, ANR and analog bypass. - A3 audio over switched Ethernet with TSN/AVB class A streams, distributed mixing in each box, and the same bypass. - A daisy-chain TDM bus was dropped before scoring: one cut isolates everything downstream.

Method. - Weights are 1–10 importance ratings, auto-normalised (Wasson's recommended method, p. 694, instead of a 100-point split, which he warns leads to "pushing points around"). - Measured criteria use linear utility (1–10) between the worst-acceptable and best values (Fig. 32.9). - Judged criteria use 1–10 scores with a written reason. - Sensitivity: ±10 % on each weight (p. 696) and 20 000 Dirichlet random weightings.

Criterion (weight)	Measure	A1	A2	A3
Fail-operational (10)	loss of commander–driver voice per 10^6 12-h missions (S6)	578	37	37
Intelligibility (9)	best STI, tracked vehicle; A1 sums 4 open mics, +6 dB noise (A)	0.27	0.44	0.44
Latency (7)	crew-to-crew worst case (S4)	0 ms	8.6 ms	8.4 ms
EMC robustness (7)	judged	4: audio-band lines, BCI audible	8: differential, CRC, concealment	7: isolated PHYs; faster edges raise the RE102 risk
Harness mass (5)	conductor-metres, 6 stations (A)	240	144	144
Growth (4)	judged	3	6	9
Cost and complexity (5)	judged	8	6	4: TSN switch, gPTP, stream reservation
Weighted total		4.81	**7.47**	7.38
Random weightings won		0.1 %	**67.7 %**	32.2 %





Recommendation: A2. - No ±10 % single-weight change flips the result. - A3 is the growth path once data and video share the network. Its advantage (no central mixer) matters less here because the bypass already removes the CCU as a single point of failure for voice.

Other viewpoints and risks (W p. 691). - If the user needs about 10 or more crew and dismount stations, star cabling grows linearly and A3's switch ports win. - The judged scores are this author's; a real study needs user ratings (W p. 692, NGT ballot).

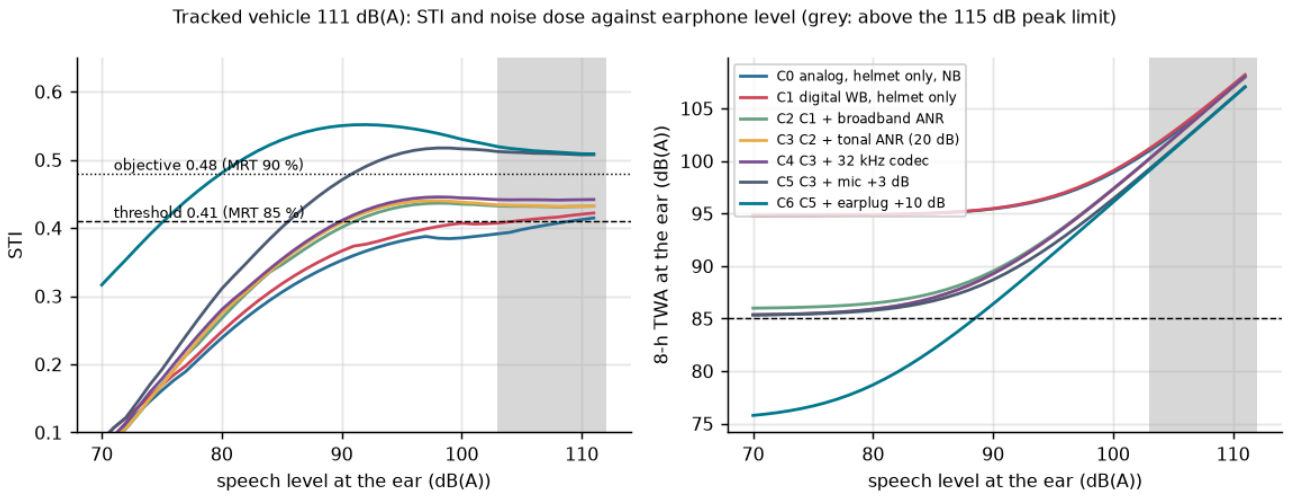
9. Modelling and simulation (W ch. 33, pp. 703–720)

9.1 M&S plan

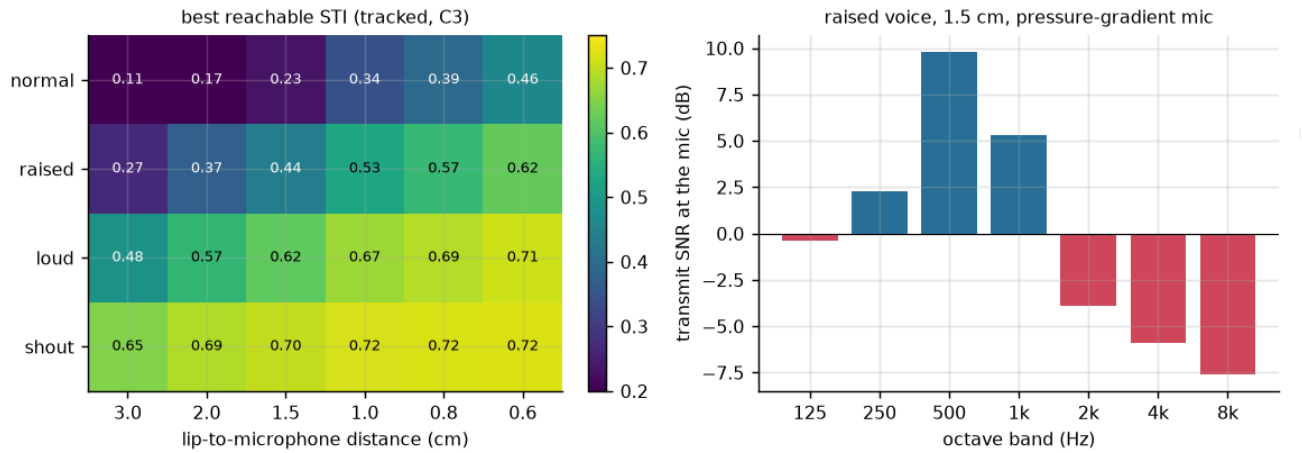
Model	Question (CTI)	Type, fidelity	Inputs (source)	Validation	Where
Speech chain + STI (S1)	STI and dose vs noise, ANR, mic, codec	deterministic, octave-band, 2nd order	spectra [R2], helmet [R17], ANR [R3], speech [R14], mic [R18], IEC constants [R11]	against the direct STIPA method (S3)	`model/icsim/chain.py`, `sti.py`
ANR (S2, S3)	convergence, stability, latency	stochastic, time-domain, sample-exact	tones from track pitch [R1], speed jitter (A), path delay	stability boundary against theory (90°), C firmware against Python	`model/icsim/anr.py`
Latency budget (S4)	design-to vs limits	deterministic worst case	converters [R37], AVB bound [R36]	Renode measures the processing term	`model/icsim/budget.py`
AoA (S5)	architecture choice	decision model	S1, S4, S6	sensitivity	`model/studies/s5_aoa.py`
RMA (S6)	mission reliability, MTBCF	parts count + RBD	MIL-HDBK-217F N2 GM [R34]	none possible without field data (A)	`model/icsim/rma.py`
28 V input (S7)	surge stress	time-domain, first order	MIL-STD-1275E envelopes [R27]	needs a bench surge test	`model/studies/s7_power.py`
Crew-box firmware	modes, timing, bypass, ANR on target	the real C code on an emulated RT1064	test image with plant model	host tests and Renode tests	`fw/`

Accreditation stance (W p. 707): validated internally, not certified. Every result file in `results/` is regenerated by `model/run_all.py` from the committed code.

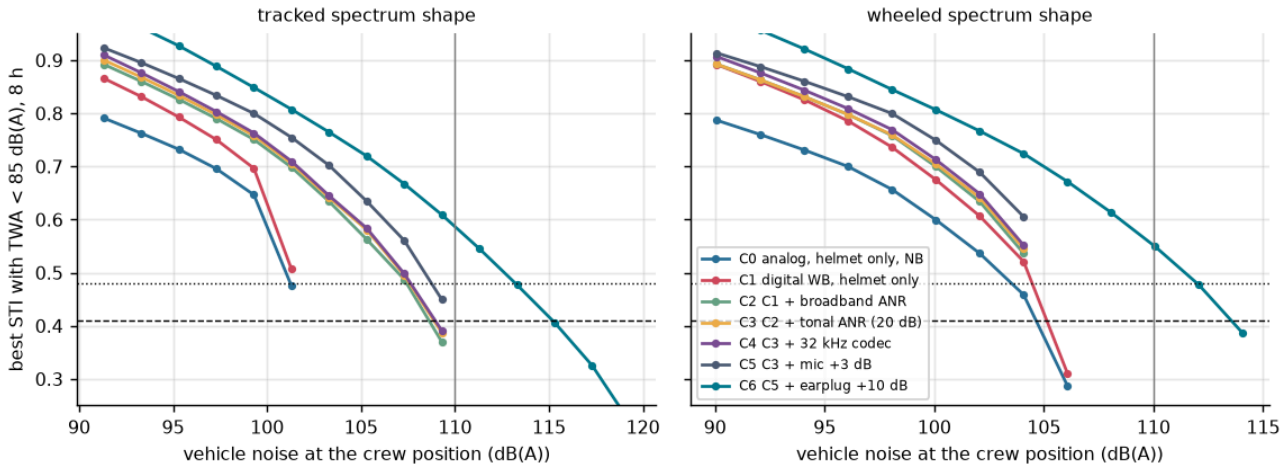
9.2 Results: where intelligibility is lost (S1)



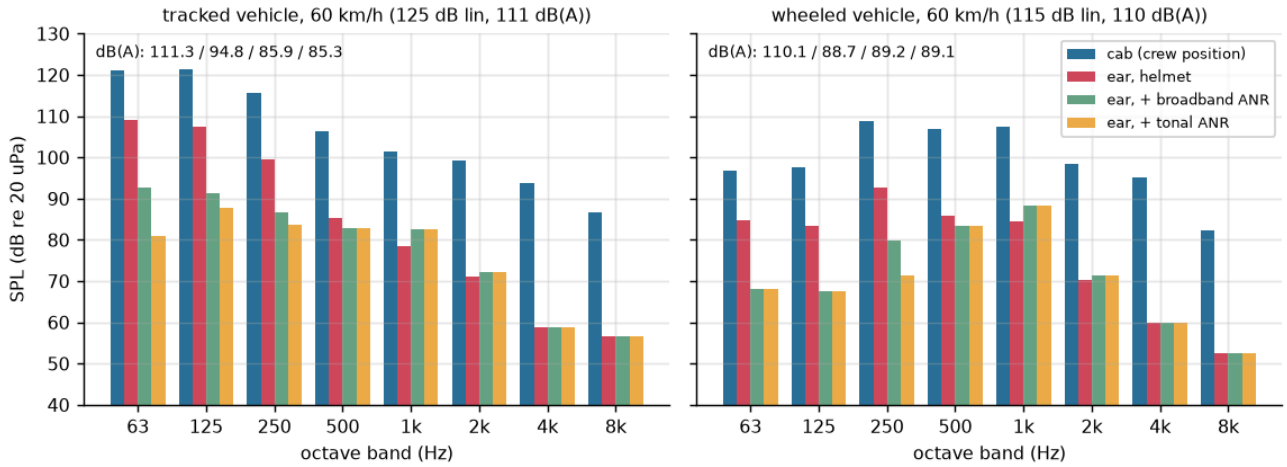
- Raising the earphone level helps until about 98 dB(A), then the STI flattens. The residual is **transmitted** noise, which rises with the speech.
- ANR lowers the level at which STI 0.41 is reached, from 100 to 91 dB(A), and that is where its dose benefit comes from.



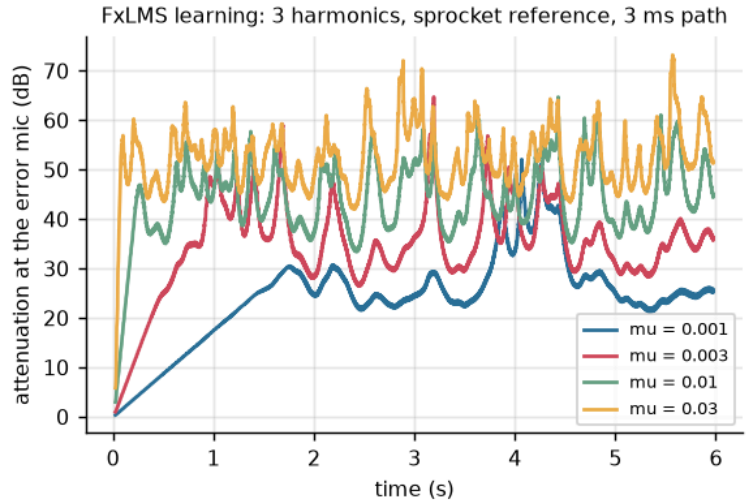
- The best reachable STI depends almost only on the microphone: lip distance and vocal effort.
- At 1.5 cm, a raised voice gives 0.44 and a loud voice 0.62.
- That is the lever for the objective, together with the MIL-STD-1472G ≥ 10 dB noise-cancelling requirement (SYS-3.3.3). The vendor's pressure-gradient curve gives 10 dB of rejection only below about 800 Hz.

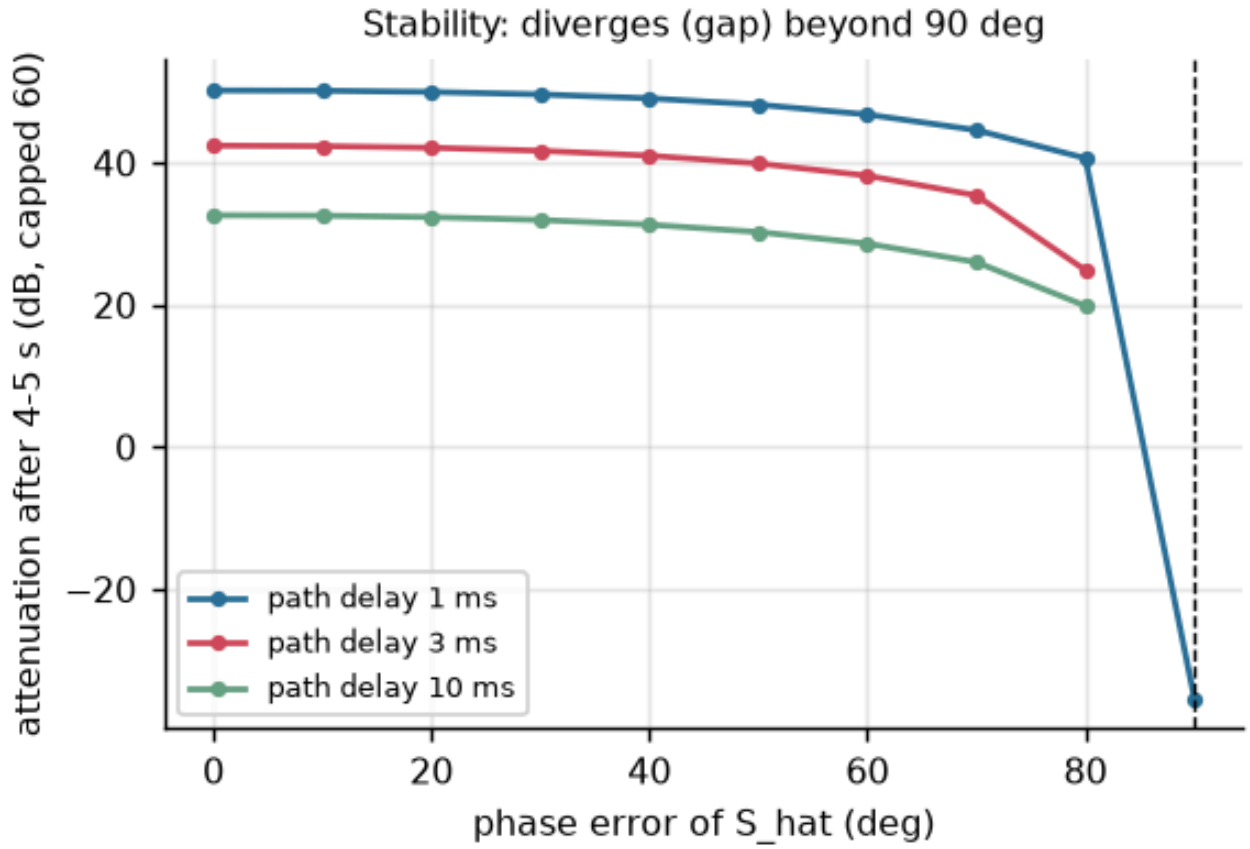


- With the dose constraint applied, every configuration loses the threshold before 110 dB(A) unless passive attenuation rises (C6).
- The wheeled vehicle is harder than its similar dB(A) suggests: its energy sits at 250 Hz–1 kHz, where the helmet is weakest (16–23 dB) and broadband ANR does little or amplifies (–4 dB at 1 kHz [R3]).



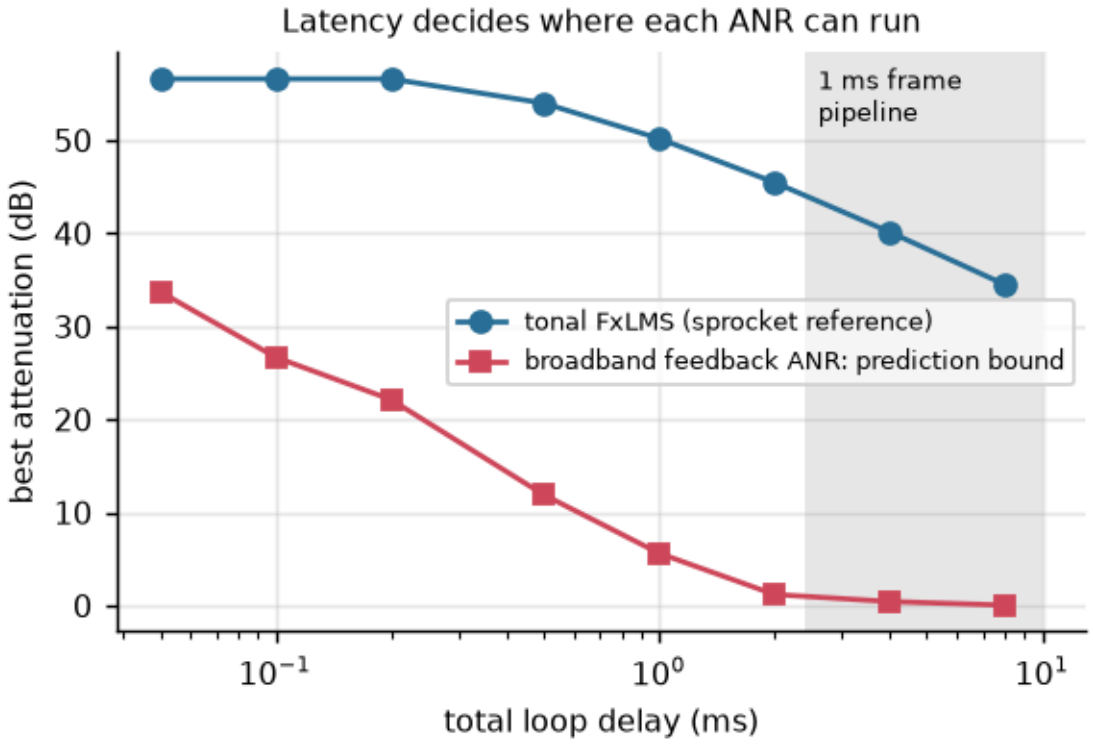
9.3 Results: adaptive ANR (S2)

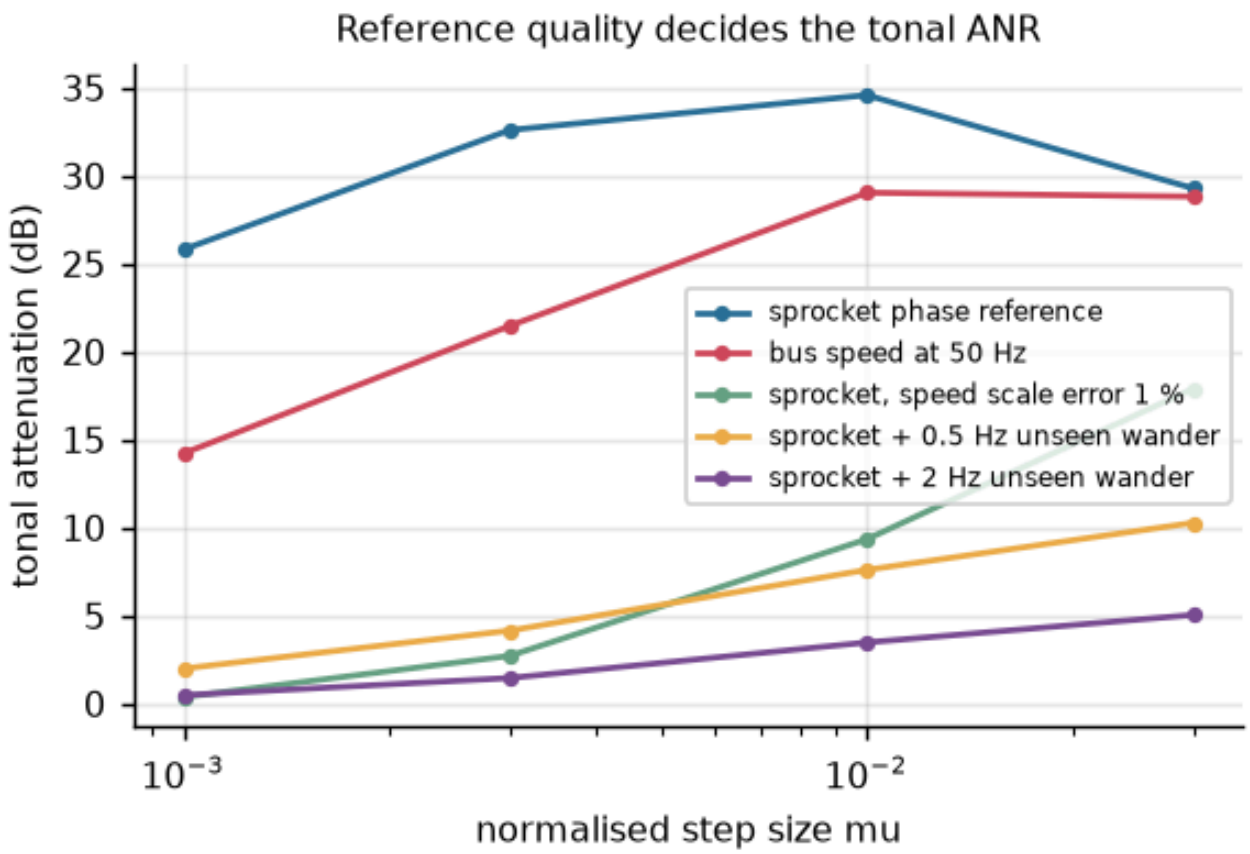




Convergence. - Normalised step size μ = 0.01 at 4 kHz reaches 90 % of its final attenuation (45 dB) in 0.73 s. - μ = 0.03 gives 54 dB but converges more slowly in this run (2.5 s), because the higher gradient noise delays the last few dB.

Stability. - Attenuation holds up to about 80° of S_hat phase error. - Beyond 90° the filter diverges (the gaps in the plot), as the filtered-x theory predicts. - The firmware therefore carries a divergence guard: error power 6 dB above its pre-ANR value for 0.2 s, or a weight-magnitude bound. A trip switches the ANR off and the box to DEGRADED.





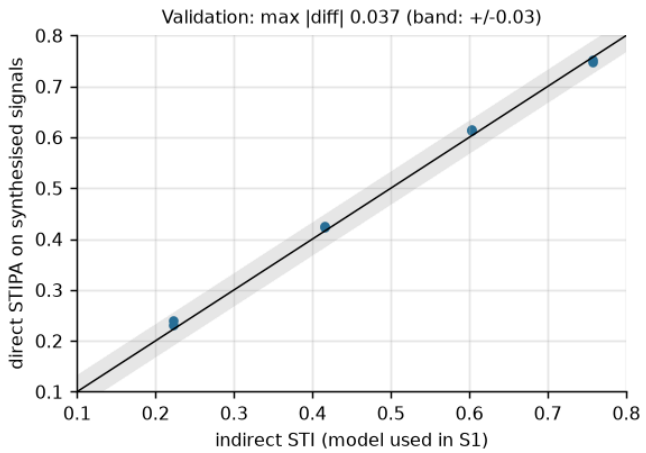
Latency. - The broadband bound (Wiener prediction of the broadband part of the ear noise over the loop delay) falls from 34 dB at 50 µs to 12 dB at 0.5 ms and 1 dB at 2 ms. - The TI converter delays at 192 kHz with filter C sum to 0.125 ms [R37], fast enough for broadband ANR. - At 16 kHz they sum to 2.4 ms, which leaves broadband ANR nothing.

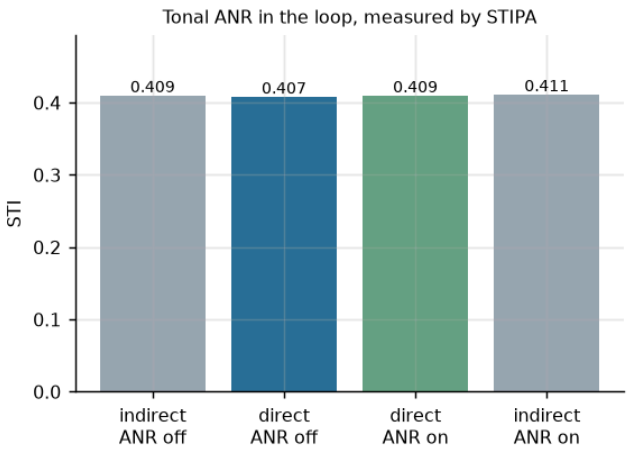
Reference quality (S2-D). The tonal ANR is limited by how well its reference follows the tones, not by delay:

Reference case	Tonal attenuation
sprocket phase	33–35 dB
bus speed at 50 Hz	29 dB
1 % scale error	9–18 dB, depending on µ
0.5 Hz of unseen wander	8–10 dB

The line width of real track tones has not been measured anywhere this study could find. **It is the first open question for the company (§15).**

9.4 Validation of the STI model (S3)





The indirect method used in S1 was checked against the direct STIPA method on synthesised signals: the modulated test signal at the S1 operating-point band levels, plus noise at five SNR shifts and two seeds. - At and above the operating point (STI ≥ 0.4), the direct values sit within **0.012** of the indirect ones. - The worst case, **0.037**, is at STI ≈ 0.1, where the STIPA random error is largest. - The mean difference is +0.011.

The tonal ANR was then run **inside** the STIPA measurement, with its error microphone hearing the speech too. The step size was scaled to the 32 kHz sample rate (μ = 0.00125, the same adaptation per second as μ = 0.01 at 4 kHz). The first run used the unscaled μ: the delayed-LMS bound was exceeded, and the "ANR" **raised** the 125 and 250 Hz bands by 3 and 6 dB. This is a design rule for the firmware: μ is specified per second, not per sample.

	Direct (STIPA, measured through the adaptive filter)	Indirect (model)
ANR off	0.407	0.409
ANR on	0.409	0.411
Band levels, on – off	125 Hz –2.1 dB, 250 Hz –0.6 dB, 500 Hz +0.02 dB, 1–8 kHz 0.0 dB	

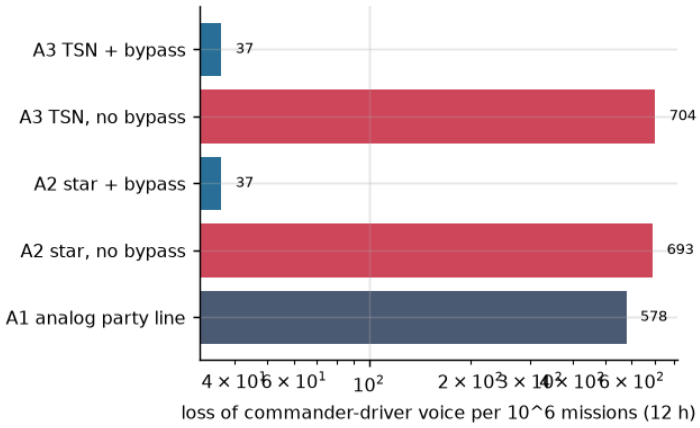
- The ANR removes energy only in the tone bands and leaves the speech bands untouched. Its error microphone hears the speech, but the reference is tonal, so it cannot cancel speech.
- **The measured STI gain is +0.002, which is negligible.** At this operating point the STI is set by the transmit side (§9.2).
- The value of the tonal ANR is noise dose and comfort, not intelligibility. The firmware therefore needs no "audio compensation" of the error signal, which saves one secondary-path filter.

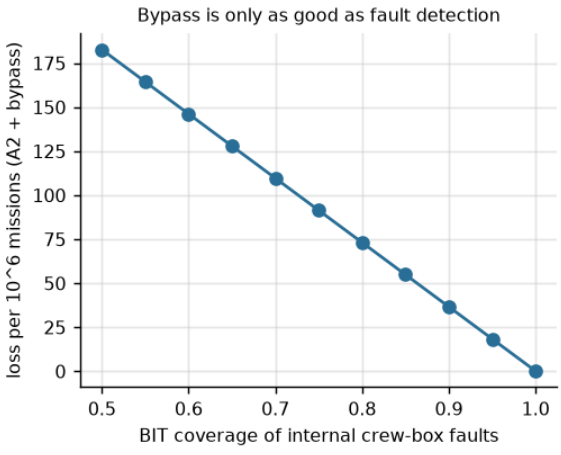
10. Reliability, maintainability, availability (W ch. 34, pp. 721–791)

Parts-count prediction (W Principle 34.28, p. 760). - MIL-HDBK-217F Notice 2 Appendix A Ground Mobile generic rates [R34] (V), with π_Q = 1 (A). - The parts lists are estimates (`model/icsim/rma.py`).

Unit	λ (per 10^6 h)	MTBF	Largest contributors
Crew box, digital path	15.2	65 700 h	ceramic chip capacitors 38 %, chip resistors 28 %, rotary switches 26 %
Crew box, emergency path	6.2	160 900 h	push switch 29 %
Crew box total	21.4	46 700 h	Ai = 0.99999 at 0.5 h MTTR (W Eq. 34.53)
CCU	27.3	36 700 h	passives 91 %

Mission reliability of commander–driver speech, 12 h (series and parallel blocks, W Eqs. 34.36–34.39). The bypass engages if the fault is detected: - with coverage 1.0 for CCU and link faults, because link supervision sees them by construction; - with coverage c (0.9, A) for internal crew-box faults.





Architecture	Losses per 10^6 missions	MTBCF
A1 analog	578	20 800 h
A2 without bypass	693	17 300 h
A2 with bypass	**37**	80 200 h
A3 with bypass	37	80 100 h

The SYS-3.2.8 objective of 0.9999 (100 per 10^6) is met with the bypass and missed without it. The coverage curve shows that BIT design, not redundancy, decides the result.

FMEA (worksheet columns of W Fig. 34.20, p. 766; severity and probability per MIL-STD-882E as in W Fig. 34.19, p. 765). Full table: `results/fmea.md`. Two rows set design work:

Row	Finding	Action
8, 9	**latent failures of the emergency path** (relay stuck energised, emergency amplifier dead) are invisible in operation	MAINT state includes a bypass self-test: de-energise the relay at standstill and loop audio through the emergency line
7	ANR divergence is a hearing hazard	guard + output clamp (CB-43, CB-46), verified on the host and in Renode

11. Qualification plan

11.1 EMC: MIL-STD-461G, Army ground [R29]

Applicable per Table V: CE102, CS101, CS114, CS115, CS116, RE102, RS103.

Test	Level (V)	Design concern
CS114	curve 3 (10 kHz–2 MHz), curve 4 (2–200 MHz); 95 / 103 dBµA flat	bulk current on the analog emergency line and headset cables: audio-band demodulation is directly audible; transformer isolation and RF filtering at the connector
CS101	126 dBµV falling to 96.5 dBµV at 150 kHz on 28 V	ripple into the codec supply; LDO PSRR budget
CS115 / CS116	5 A, 2 ns rise / 10 A damped sinusoid	RS-485 transceiver protection; CRC and concealment make a corrupted frame inaudible (IT04)
RS103	50 V/m, 2 MHz–18 GHz (40 GHz optional)	own transmitters next to the crew box: the worst case in practice
RE102	44 → 89 dBµV/m (Army)	RS-485 edge rate; slew-limited transceivers
CE102	94 dBµV basic curve (28 V)	DC-DC converter input filter

11.2 Environment: MIL-STD-810H [R30]–[R32]

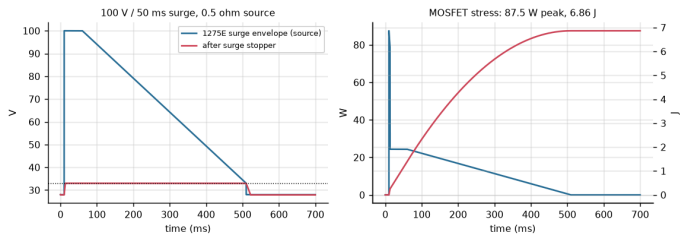
Method	Parameters	Status
501.7 high temperature	A1 hot-dry: 32–49 °C ambient, up to 71 °C induced (Table 501.7-I)	V
502.7 low temperature	C2 cold: –37 to –46 °C (Table 502.7-I)	V
514.8 vibration	ground-vehicle categories; a fielded vehicle intercom uses tracked/wheeled profiles (810G category 20)	vendor practice [R33]
516.8 shock	functional and crash-hazard shock	parameters not verified (A)
507.6 humidity	to be specified	–
509.7 salt fog	to be specified	–
510.7 sand and dust	to be specified	–
512.6 immersion	to be specified	–

The 509.7, 510.7 and 512.6 parameters are not verified; they come from the procuring specification.

11.3 Power: MIL-STD-1275E [R27]

- Steady state: 20–33 V (5.1.1.1).
- Surge: 100 V for 50 ms, 0.5 Ω, 60 J (Table I).
- Spikes: ±250 V for 70 µs, 2 J (Fig. 7).
- IES: 12 V for 1 s. Cranking: 16 V for 30 s (5.1.2).
- Reverse polarity: the equipment protects itself (4.1).

Input-stage design, found by analysis S7:



1. **Reverse polarity:** an ideal-diode MOSFET.
2. **TVS for spikes only.** Its stand-off must be above 100 V. A TVS clamping at 36 V would absorb about 230 J during the 1275E surge (128 A for 50 ms) and fail.
3. **Linear surge stopper** clamping the output at 33 V, with a 1 A current limit. The MOSFET takes **6.9 J and an 87 W peak** over the surge. That sets its SOA requirement, and its V_DS rating must exceed the TVS clamp voltage.
4. **Wide-input DC-DC, 9–36 V.** It covers IES and cranking with no hold-up. Without it, riding through the 1 s IES from 28 V down to 16 V would need **22.7 mF** at 6 W. A 50 ms dropout needs 0.9 mF.

12. Software assurance: DO-178C-style plans and evidence

This is the style of DO-178C, not compliance: no certification authority, no independence, no qualified tools. The plans are condensed into this section.

Plan or objective	What is done here	Evidence
Software requirements (SRS)	CB-01 … CB-47, derived from SYS-3.x	`fw/SRS.md`
Design standard	pure C99 core library, no heap, no OS calls, caller-owned state; Zephyr glue separate	`fw/lib/icom`, `fw/app`
Coding standard	MISRA C:2012 via cppcheck 2.13; one documented deviation (D1: rule 2.5, protocol constants in the public header)	`fw/scripts/misra.sh`, `fw/docs/misra.md`
Structural coverage	100 % statement, branch and **MC/DC** of the core library (GCC 14 `-fcondition-coverage`)	`fw/scripts/test-host.sh`: 500/500 lines, 274/274 branches, 274/274 MC/DC condition outcomes
Requirements-based tests	1283 host checks; IT01–IT08 in Renode on the real firmware image	`fw/tests`
Traceability	every CB requirement to a test; checked in CI	`fw/tools/trace.py`, `fw/docs/traceability.md`
Configuration management	git; CI rebuilds the evidence	`.github/workflows/ci.yml`
Not done	independence, tool qualification, object-code coverage, hardware/software integration on the real board, WCET	§14

13. Verification and validation (W ch. 13, pp. 270–292; ch. 28, pp. 599–622)

13.1 Strategy

- **Verification** shows compliance with the specification; **validation** shows the user's need is met (W Principles 13.3/13.4, p. 275).
- Use the fewest, cheapest methods that prove compliance (W Principle 13.11, p. 283). Design verification must cover robustness to failure (p. 280).

Requirement	Method	Level	How
SYS-3.2.1 STI	A now; T later	system	S1 model validated by S3; later a head-and-torso simulator in a noise chamber at the recorded vehicle spectrum, STIPA through the real chain
SYS-3.2.2 dose	A, T	system	S1; dosimeter in an ear simulator
SYS-3.2.4/5 latency	A, T	crew box, system	S4; time-stamped loop-back at the converters
SYS-3.2.7 failover	T	crew box (Renode now), system later	IT02, IT03, IT07, IT08; then pull cables and the CCU on the bench
SYS-3.2.8 reliability	A	system	S6 parts count, then FRACAS from prototypes
SYS-3.4.1 power	A, T	unit	S7; 1275E generator tests
SYS-3.5.x EMC/ environment	T	unit	accredited laboratory

13.2 Integration (W Figs. 28.2/28.4)

1. **IP1:** crew-box firmware on the host (done) and in Renode with a CCU emulator (done).
2. **IP2:** crew box with headset on the bench, simulated bus.
3. **IP3:** two crew boxes with bus and CCU.
4. **IP4:** full crew complement with emulated radios.
5. **IP5:** in a hull or mock-up, with vehicle power and the data bus.

Threaded test cases (W Principle 28.3), for example "commander talks to driver at 85 / 100 / 110 / 115 dB(A)": - verifies STI, latency and sidetone in one run; - is then repeated with box 3 failed, a bus cut and the CCU off.

13.3 Evidence produced in this repository

Host. - `fw/scripts/test-host.sh`: 1284 checks pass, with 100 % of 500 lines, 274 branches and 274 MC/DC condition outcomes (GCC 14). - `fw/scripts/misra.sh`: no MISRA C:2012 findings apart from deviation D1. - `fw/tools/trace.py`: 34 of 34 CB requirements traced, 0 errors. - `model/tests`: 16 tests pass.

Renode (`fw/scripts/test-sim.sh`, the test image on the MIMXRT1064-EVK model, with `fw/sim/rig.py` as the CCU):

Test	Requirement	Result (virtual time)
IT01 Boot to NORMAL	CB-20, CB-27	NORMAL 1.35 ms after boot; relay energised
IT02 CCU loss to BYPASS	CB-22, CB-27, CB-04	BYPASS **20.3 ms** after the last frame; relay dropped at 21.3–22.3 ms (varies by run with the tick phase); telemetry continues
IT03 Recovery and no chatter	CB-23	0 transitions in 3 s at 50 % frame loss; NORMAL **2000.0 ms** after the link became good
IT04 Corrupted frames	CB-12, CB-15	1 frame in 10 corrupted for 2 s: 203–204 CRC errors, 203 frames lost (a frame corrupted in the last tick is counted when the next one arrives), no mode change
IT05 ANR on target	CB-40, CB-41, CB-05	120 dB on the plant's tones; worst `icom_box_tick` 98 838 cycles = **165 µs**
IT06 Wrong S_hat caught	CB-28, CB-46, CB-43	DEGRADED **599 ms** after moving off with a 150° error; largest earphone sample 16 384 = the 0.5 FS clamp; relay stays energised
IT07 Critical BIT latches	CB-24	BYPASS in < 10 ms; still BYPASS 3 s after the fault cleared
IT08 Processor stall	CB-27	relay dropped **1.4–2.1 ms** after the stall (3 runs)

Notes on the results: - **IT05.** The 120 dB holds only for the plant's exactly periodic tones with an exact path estimate. The realistic figures are the S2 ones (§9.3). The cycle count is Renode's virtual time, not silicon timing. - **IT04** found a real defect: the first version counted each corrupted frame twice (406 lost for 203 CRC errors). It is fixed in `link.c`, and the fix is recorded in `fw/SRS.md`.

13.4 Validation

- Validation needs real crews doing scripted use cases (fire orders, driver commands, radio relay) in a running vehicle or a noise chamber, scored by task errors (W Principle 13.14, p. 289).
- Not possible in a portfolio project. It is the main piece an independent test authority would add (W Principle 13.15).

14. What is sourced, what is assumed, and the limits

Sourced: - noise spectra [R1][R2]; - helmet attenuation [R17]; - ANR insertion gain [R3]; - speech spectra [R14]; - STI constants [R11][R13]; - MRT↔STI table, dose and communication limits [R4][R5]; - delay tolerance [R20]–[R24]; - 1275E, 461G and 810H (temperatures) [R27]–[R32]; - 217F rates [R34]; - converter delays [R37].

Assumed (A): - lip-to-mic distance 1.5 cm; - speech duty 30 % of the shift and an 8-h exposure; - the crest factor of 12 dB; - the 63 Hz helmet value; - the tonal-share method; - the track-tone line width and speed jitter; - the parts lists and $\pi_Q = 1$; - BIT coverage 0.9; - the mission length of 12 h; - the harness layout; - the judged AoA scores; - the crew-to-crew latency limit, which is a proxy.

Limits: - **Emulated hardware.** - The RT1064 runs in Renode. The SAI and audio codec are not modelled, so the test image replaces the codec with an acoustic plant model: track tones from the bus speed plus the delayed anti-noise. - The bus is a UART at the simulator's speed, not a timed RS-485 line. - Cycle counts are Renode's, not the silicon's. - **The CCU is not built.** Its behaviour exists only as the Renode rig and as model assumptions. - **The STI results are model results.** They are validated against the direct method on synthetic signals, not against listeners or a measured vehicle. - **The broadband ANR curve** is one published prototype headset [R3], not a current product. - **No certification.** This is DO-178C-style evidence only (§12). - **Vehicle data is from published papers** about other vehicles. The company's own vehicles and headsets would change every number in §7.1.

15. Open engineering questions, and why they matter to a vehicle-intercom maker

1. **How wide are real track tones, and can a sprocket-phase signal be had from the vehicle?** This decides whether speed-referenced tonal ANR gives about 30 dB or about 8 dB (§9.3). It is cheap to measure: one recording at the error microphone together with the sprocket pick-up.
2. **Transmit-side SNR at 2–8 kHz.** The model says the microphone, not the ear, limits STI above about 105 dB(A). Measuring the noise rejection of the company's boom microphone and shield in the 2–8 kHz bands would confirm or refute that directly.
3. **Dose at ≥ 110 dB(A).** No intercom-only design meets an 8-h TWA of 85 dB(A) at 111 dB(A) with a helmet at the MIL-DTL-44117B attenuation level. The system answer includes the earplug or the exposure time, so it belongs in

the user's concept of use.
4. **Analog vs digital sidetone** (MIL-STD-1472G "in phase") and the converter path for broadband ANR: both point at the codec's analog bypass and low-latency filter options.
5. **BIT coverage of the crew box** is the parameter that decides the fail-operational number (S6). It should be designed and measured (fault-injection campaign), not assumed.

16. References

Wasson

[W] C. S. Wasson, *System Engineering Analysis, Design, and Development: Concepts, Principles, and Practices*, 2nd ed. Hoboken, NJ: Wiley, 2016. Chapters 4–7, 13, 19–22, 26–28 and 31–34 were read; the pages are cited in the text.

Vehicle noise and hearing protection

[R1] D. G. Zimcik, R. M. Provencher, B. Lecheminant, and D. McNamara, "Measurement of noise in armoured personnel carriers," NATO RTO-MP-AVT-110, paper 9, 2004.

[R2] K. Buck, P. Hamery, and V. Zimpfer, "The European regulation 2003/10/EC and the impact of its application to the military noise exposure," in *Proc. ICA 2010*, Sydney, paper 495.

[R3] S. J. van Wijngaarden and S. James, "Protecting crew members against military vehicle noise," NATO RTO-MP-AVT-110, keynote KN1, 2004.

[R17] MIL-DTL-44117B, *Helmet, Combat Vehicle Crewman, Ballistic Shell*, Table I.

[R18] Vendor datasheet, pressure-gradient dynamic boom microphone, 150 Ω (far-field crossover 2.5 kHz, –6 dB/octave). The name is withheld under the repository's naming rule and kept in the author's notes.

Standards

[R4] MIL-STD-1474E, *Noise Limits*, U.S. DoD, 15 Apr. 2015.

[R5] MIL-STD-1472G w/Change 1, *Human Engineering*, U.S. DoD, 2019. 1472H (2020) exists but was not read.

[R9] ISO 9921:2003, *Ergonomics: Assessment of speech communication*.

[R10] IEC 60268-16:2020 ed. 5, *Objective rating of speech intelligibility by speech transmission index*. Preview only; constants via [R11]–[R13].

[R24] ITU-T Rec. G.114, *One-way transmission time*, 2003.

[R27] MIL-STD-1275E, *Characteristics of 28 Volt DC Input Power to Utilization Equipment in Military Vehicles*, 2013.

[R29] MIL-STD-461G, *Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment*, 2015.

[R30] MIL-STD-810H method list (secondary).

[R31] MIL-STD-810H, Method 501.7.

[R32] MIL-STD-810H, Method 502.7.

[R34] MIL-HDBK-217F Notice 2, *Reliability Prediction of Electronic Equipment*, 1995, Appendix A.

Speech intelligibility and STI

[R7] NTi Audio, STI know-how page (rating bands).

[R11] Architectural Institute of Japan, STI benchmark problems B-32 / A-21 (IEC 60268-16:2020 constants).

[R12] MathWorks, "Speech transmission index IEC 60268-16 edition 5 compliance."

[R13] P. Rajmic, J. Schimmel, and Š. Cieslar, "A MATLAB toolbox for computation of speech transmission index," arXiv:2510.03825, 2025.

[R14] ANSI S3.5-1997 speech spectra, via the R package "SII" documentation.

[R15] P. Bottalico et al., "Evaluation of the starting point of the Lombard effect," *Acta Acust. united Acust.*, 103(1), 2017.

[R38] P. C. Loizou and G. Kim, "Reasons why current speech-enhancement algorithms do not improve speech intelligibility," *IEEE Trans. Audio, Speech, Lang. Process.*, 19(1), 2011. This is why single-channel noise suppression is not counted on for STI here (from memory, not re-read in this session).

Delay tolerance

[R20] M. A. Stone and B. C. J. Moore, "Tolerable hearing aid delays. I," *Ear Hear.*, 20(3), 1999.

[R21] M. A. Stone and B. C. J. Moore, "Tolerable hearing aid delays. II," *Ear Hear.*, 23(4), 2002.

[R22] M. A. Stone and B. C. J. Moore, "Tolerable hearing aid delays. III," *Ear Hear.*, 24(2), 2003.

[R23] M. A. Stone and B. C. J. Moore, "Tolerable hearing aid delays. IV," *Ear Hear.*, 26(2), 2005.

Equipment, networks and ANR theory

[R33] Vendor datasheet of a fielded digital vehicle intercom (environmental qualification table). The name is withheld; kept in the author's notes.

[R36] Audio Video Bridging stream-reservation class A bound of 2 ms over 7 hops (secondary; IEEE 802.1BA not read).

[R37] Texas Instruments, *TLV320AIC3204 Application Reference Guide*, SLAA557, 2012, Tables 2-11 to 2-13 and 2-19 to 2-21 (converter filter group delays).

[R39] S. M. Kuo and D. R. Morgan, *Active Noise Control Systems: Algorithms and DSP Implementations*, Wiley, 1996. Filtered-x LMS and its ±90° phase condition (background, not re-read).

Where the evidence is. The verification status of every number, the URLs, and the record of what could not be fetched (DTIC was offline) are in `docs/sources.md`.